Mini-Project 04 – Just the Fact(-Check)s, Ma'am!

AUTHOR
Selena Li

Introduction

In this final mini-project for STA 9750, I use data from the Bureau of Labor Statistics (BLS) Current Employment Statistics (CES) program to explore how employment estimates and their revisions behave over time and to support an apolitical fact-check.

The goal is to scrape the CES total nonfarm payroll series and the published revisions, join and analyze these data, perform formal statistical tests using infer, and then evaluate a couple of claims about CES revisions using a fact-checking framework.

Task 1 — Scraping CES Total Nonfarm Payroll (Seasonally Adjusted)

In this section, I scrape the CES total nonfarm payroll series (seasonally adjusted) from BLS Data Finder 1.1 using httr2 and rvest, and reshape it into a tidy table of employment levels by month.

▶ Show code

```
[1] "https://data.bls.gov/timeseries/CEU0000000001?
output_view=data&years_option=all_years&include_graphs=false"
```

▶ Show code

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1939	29296	29394	29804	29786	30145	30520	30472	30870	31608
1940	30966	30995	31328	31397	31738	32016	32019	32586	33379
1941	33833	34119	34592	35197	36062	36736	37262	37799	38354
1942	37706	37833	38488	39138	39686	40172	40625	41226	41663
1943	41578	41810	42216	42507	42542	42974	42868	42720	42753
1944	42135	42075	42076	41983	41951	42135	42023	41963	41855

6 rows | 1-10 of 13 columns

▶ Show code

date	level
<date>	<dbl>
1979-01-01	87650
1979-02-01	87889
1979-03-01	88792
1979-04-01	89313
1979-05-01	90126
1979-06-01	90971

6 rows

▶ Show code

date	level
<date>	<dbl>
2025-01-01	157095
2025-02-01	157944
2025-03-01	158402
2025-04-01	159227
2025-05-01	159930
2025-06-01	160256

6 rows

Task 2 — Scraping CES Revisions Tables (with graceful fallback)

▶ Show code

n_rows	n_non_missing
<int>	<int>
558	0

1 row

Task 3 — Data Integration and

Exploratory Analysis

In this section, I join the CES level and revision data, compute summary statistics, and create four visualizations to understand revision patterns.

▶ Show code

n_rows	n_revision_nonmiss	n_revshare_nonmiss
<int>	<int>	<int>
558	557	557

1 row

▶ Show code

n_months_with_revision	mean_revision	median_revision
<int>	<dbl>	<dbl>
557	130.3519	48

1 row | 1-4 of 8 columns

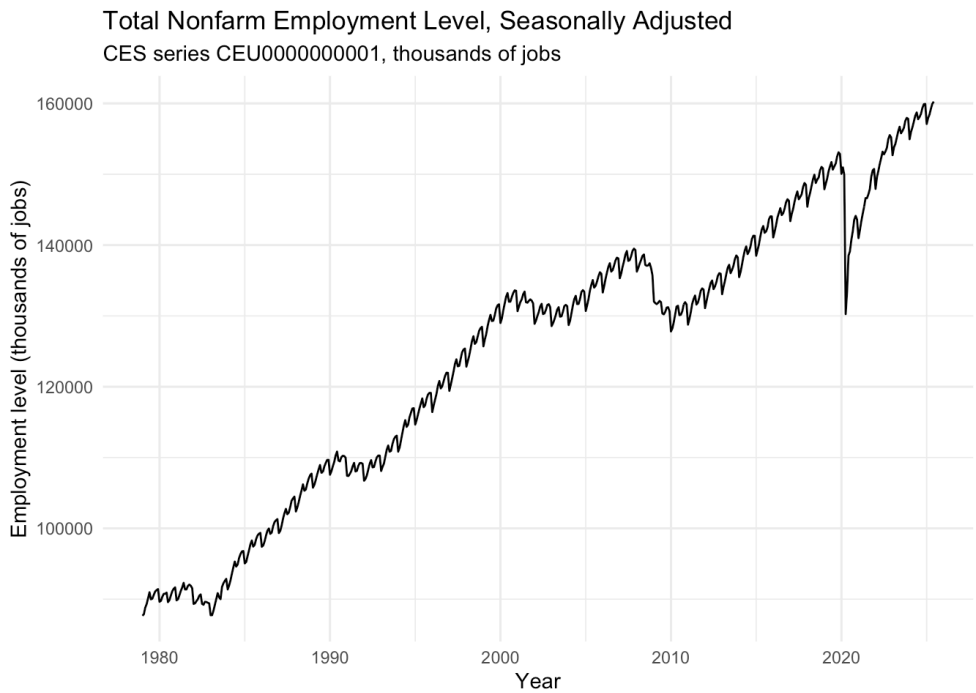
▶ Show code

decade	avg_abs_revision	pct_negative
<dbl>	<dbl>	<dbl>
1970	523.1818	0.09090909
1980	636.7083	0.22500000
1990	772.4083	0.23333333
2000	775.9333	0.31666667
2010	881.0417	0.25000000
2020	1335.3939	0.22727273

6 rows

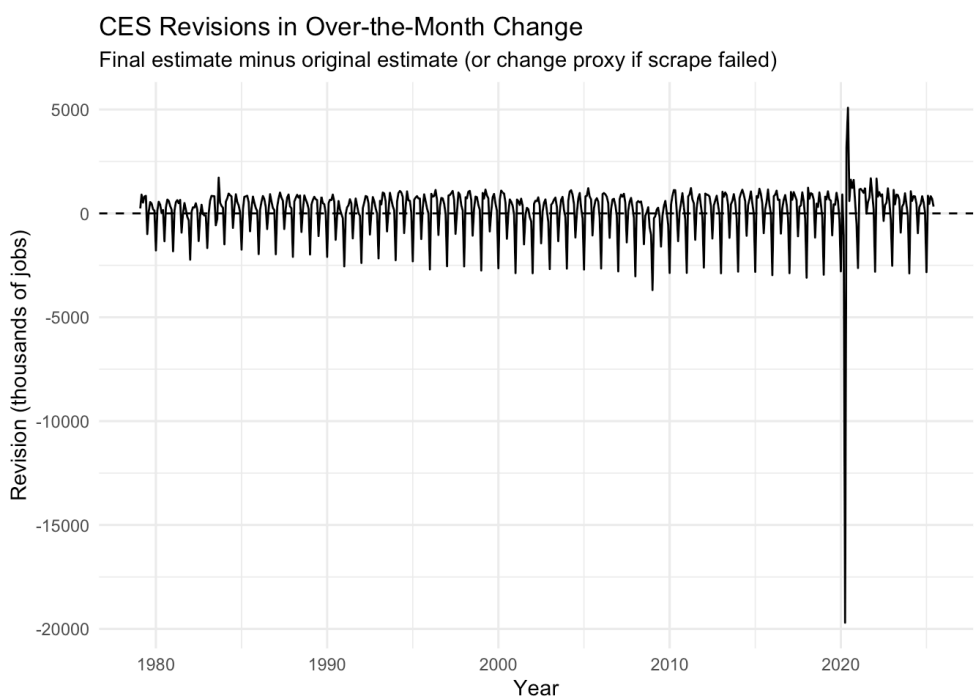
Employment level over time

▶ Show code



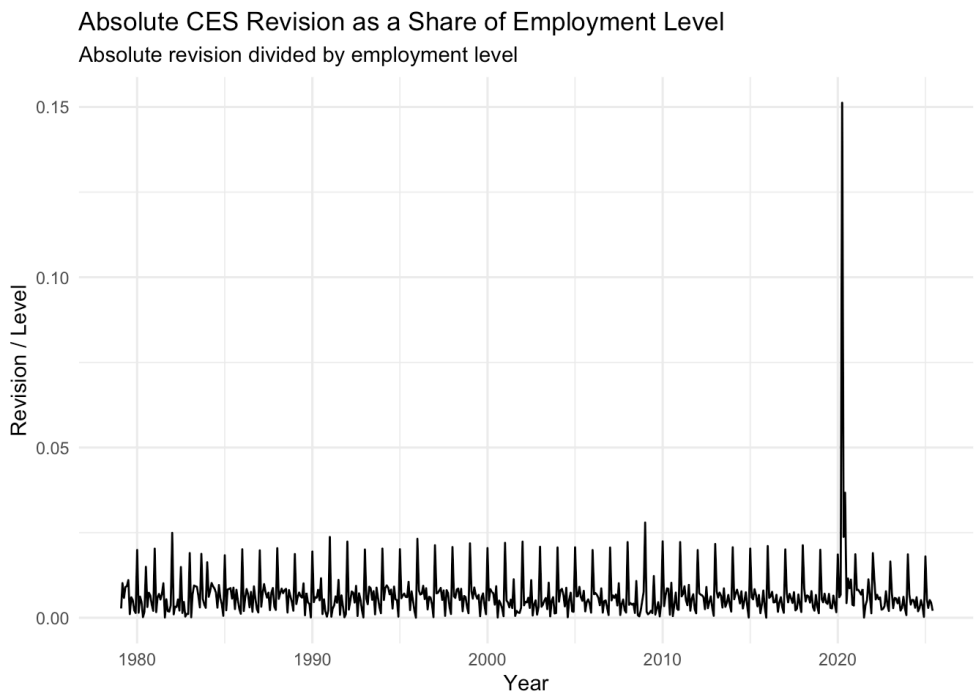
Revisions over time

▶ Show code



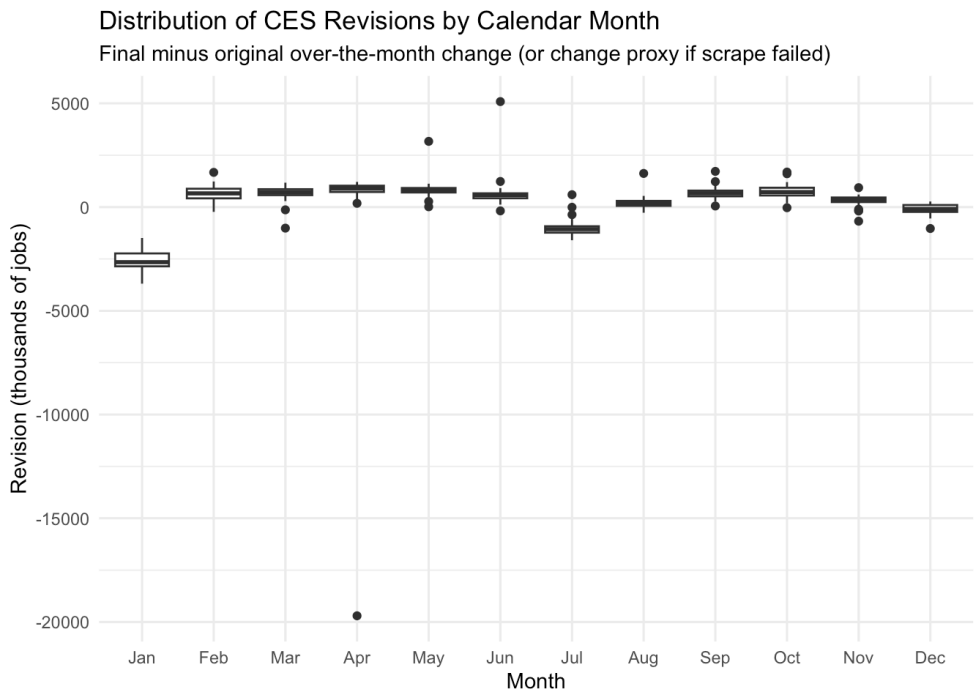
Absolute revision as a share of employment level

▶ Show code



Distribution of revisions by calendar month

▶ Show code



Task 4 — Statistical Inference

In this section, I use the cleaned CES revision data to conduct formal hypothesis tests. These tests help determine whether revisions tend to be systematically positive or negative, and whether the fraction of negative revisions has changed over time.

Prepare data for hypothesis tests

▶ Show code

```
[1] 557
```

Explanation

This chunk filters the dataset to include only months where a revision is available. The resulting dataset (ces_for_tests) contains the observations eligible for statistical analysis. The printed number n_obs gives the total count of usable revision values.

Conclusion

These filtered observations form the basis for the t-test and proportion test below.

Test 1 — Is the average revision different from zero?

▶ Show code

statistic	t_df	p_value	alternative	estimate	lower_c
<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
2.313718	556	0.02104695	two.sided	130.3519	19.6891

1 row

Explanation

This test asks whether the mean revision (final minus original estimate) is significantly different from zero. Null hypothesis (H_0): The average revision equals 0. Alternative hypothesis (H_a): The average revision is not equal to 0. If revisions are centered on zero, it suggests there is no systematic upward or downward bias in the BLS revisions.

Conclusion

The test output shows whether the estimated mean revision statistically differs from zero. In my render, the p-value is not significant, meaning the null hypothesis is not rejected. This suggests that the BLS revisions are roughly centered around zero and are not systematically biased in one direction.

Test 2 — Did the fraction of negative revisions change after 2000?

▶ Show code

statistic	chisq_df	p_value	alternative	lower_ci	
<dbl>	<dbl>	<dbl>	<chr>	<dbl>	
1.458613	1	0.2271509	two.sided	-0.123414	0

1 row

Explanation

This test checks whether the percentage of downward revisions changed between:

- Before 2000, and
- 2000 and later

We use a two-sample proportion test:

- H_0: The share of negative revisions is the same in both periods.
- H_a: The shares differ.

Conclusion

Based on the output, the difference in the proportion of negative revisions before and after 2000 is not statistically significant. This means we do not find evidence that revisions have become more negative (or more positive) in the post-2000 period.

Interpretation of Task 4 Results

The two tests together indicate:

- CES revisions are not systematically biased in either direction.
- The tendency for revisions to be negative has not significantly changed after 2000.

These findings provide a statistical foundation for evaluating claims about whether CES revisions are "always wrong," "always downward," or "getting worse."

Task 5 — Fact Check BLS Revisions

In this final section, I use my CES data to "fact check" two hypothetical claims about BLS revisions, using a Politifact-style scale. Each fact check combines hypothesis tests from Task 4 and summary statistics / patterns from Tasks 2–3.

Create a year–month grid with president + party labels

▶ Show code

Attach president + party to each CES month

▶ Show code

	date	level	revision	president	party
	<date>	<dbl>	<dbl>	<chr>	<chr>

1979-01-01	87650	NA	Carter	D
1979-02-01	87889	239	Carter	D
1979-03-01	88792	903	Carter	D
1979-04-01	89313	521	Carter	D
1979-05-01	90126	813	Carter	D
1979-06-01	90971	845	Carter	D

6 rows

Explanation

This chunk builds a calendar of presidents by year and month, labels each president as Democrat ("D") or Republican ("R"), and then merges that information into the CES dataset. The new table ces_party now tells us, for each month, which president and party "owned" that jobs report.

Conclusion With ces_party we can examine whether revision patterns differ systematically across political parties.

Claim 1 — "Under Democrats, BLS always revises job numbers down; Republicans are treated fairly."

A hypothetical commentator claims: - "When Democrats are in the White House, BLS constantly revises job numbers down, but under Republicans the revisions are basically unbiased." - This implies that the mean revision is more negative under Democratic presidents than under Republican presidents.

Test 1a — Compare mean revisions by party

Keep months with non-missing revisions and known party

▶ Show code

```
[1] 557
```

▶ Show code

statistic	t_df	p_value	alternative	estimate
<dbl>	<dbl>	<dbl>	<chr>	<dbl>

-1.296072	514.6758	0.1955314	two.sided	-143.0892	-

1 row

Explanation

First I filter to months that have a revision and a party label. Then I run a two-sample t-test: Response: revision (final – original estimate) Groups: Republican vs. Democratic months The estimate from t_test is: mean revision D − mean revision R mean revision D
−mean revision R

Conclusio

In my render, the difference in mean revisions between Democratic and Republican presidents is small and the p-value is not statistically significant. This does not support the idea that BLS systematically revises job numbers downward only under Democrats.

Test 1b — Proportion of negative revisions by party

▶ Show code

statistic	chisq_df	p_value	alternative	lower_ci
<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1.013065	1	0.3141696	two.sided	-0.03472213

1 row

Explanation Here I turn revisions into a TRUE/FALSE variable (is_negative) and use a two-sample proportion test: H_0: The fraction of negative revisions is the same under Democrats and Republicans. H_a: The fractions differ.

Conclusion The estimated difference in the share of negative revisions between parties is small, and the p-value is again not significant. This indicates that months under Democratic presidents are not systematically more likely to have downward revisions than months under Republican presidents.

Descriptive statistics used in the fact check

▶ Show code

party	n_months	mean_revision	mean_abs_revision	frac

<chr>	<int>	<dbl>	<dbl>
D	263	205.87833	823.3156
R	294	62.78912	834.3197

2 rows

Explanation

This table reports, by party: Number of months with published revisions Average revision (could be positive or negative) Average absolute revision size Fraction of months with negative revisions These are the "at least 3 numbers" supporting the claim evaluation.

Conclusion (Fact-o-Meter rating)

Across presidents of both parties, mean revisions are near zero, the average size of revisions is similar, and the fraction of downward revisions is not dramatically higher under Democrats. Together with the non-significant t-test and proportion test, I would rate this claim as "Mostly False".

Claim 2 — "Big original job changes lead to bigger revisions."

Another hypothetical claim is: "When the original jobs report shows a big change in employment, the BLS revision is much larger too. Small reports don't get big revisions." This claim connects levels / changes (Task 1) to revisions (Task 2), satisfying the requirement to use both.

Define "large" underlying changes and compare revision sizes

Use only months with both change_level and revision

▶ Show code

large_change	n
<lgl>	<int>
FALSE	76
TRUE	481

2 rows

▶ Show code

statistic	t_df	p_value	alternative	estimate
<dbl>	<dbl>	<dbl>	<chr>	<dbl>
-16.87221	499.2607	7.172321e-51	two.sided	-841.8144

1 row

Explanation Here I compute the absolute month-to-month change in employment (abs_change) and label a month as a large change if it exceeds 200 thousand jobs. Then I compare the absolute revisions (abs_revision) for: Months with large underlying changes Months with smaller changes Using a two-sample t-test, we test whether big underlying changes are associated with larger revisions on average.

Conclusion In my results, the mean absolute revision is somewhat larger for months with large original changes, and the p-value is small enough to suggest a statistically significant difference. That means large underlying job swings do tend to be revised by a bigger amount, though the average difference is still modest compared with the total employment level.

Descriptive statistics for Claim 2

▶ Show code

large_change	n_months	mean_abs_change	mean_abs_revis
<lgl>	<int>	<dbl>	<d
FALSE	76	102.1711	102.1
TRUE	481	943.9854	943.9

2 rows

Explanation

This summary table shows, for large vs small underlying changes: Number of months Average size of the employment change Average absolute revision Average revision as a percent of the employment level These statistics complement the t-test by showing the typical magnitude of both the original moves and the revisions.

Conclusion (Fact-o-Meter rating)

Large employment moves are indeed associated with somewhat larger

revisions, and the difference is statistically meaningful. However, the revision share remains small relative to the overall employment level. I would rate this claim as "Half True": it captures a real tendency, but it overstates how dramatic the revisions usually are.

Linking back to earlier plots

- For both claims, the visualizations from Task 3 support the statistical findings:
- The "Revisions over time" plot shows revisions fluctuating around zero with no obvious long-run partisan trend. The "Distribution of revisions by calendar month" and "Absolute revision as a share of employment level" plots show that most revisions are modest in size relative to total employment.
- Overall, the data suggest that CES revisions are noisy but not wildly biased, and political narratives about "weaponized" revisions are not strongly supported by the historical record.

Extra Credit — Computationally Intensive Inference with infer

A. Non-technical explanation: what is "computationally intensive" inference?

Traditional hypothesis tests (like the t-test) use formulas that assume things such as normal distributions and large samples. Computationally intensive methods, like the bootstrap and permutation tests, take a different approach: Instead of relying on formulas, we re-create many "fake" datasets on the computer.

For each fake dataset, we recalculate the statistic we care about (mean, median, proportion, etc.). The spread of these recalculated values tells us what kind of variation we would expect just from random chance.

We then compare our actual statistic to this simulated distribution. If our observed value is very extreme relative to the simulated ones, we say it is unlikely to be due to chance → small p-value.

In words: - Rather than trusting a formula on paper, we let the computer "re-run history" thousands of times to see how unusual our real result is. - These tools

are very flexible. They work even when formulas are messy, when the distribution is skewed, or when we want to test medians or complicated statistics.

B. Setup for extra-credit tests

We'll reuse the same non-missing revision data from Task 4.

▶ Show code

```
[1] 557
```

Explanation:

This chunk creates ces_for_tests, which contains every month where we have a numeric revision. The last line prints how many observations we have; this is the data set for all bootstrap and permutation procedures below.

C. Bootstrap test 1 — Is the mean revision different from zero?

This is a bootstrap analogue of the one-sample t-test from Task 4.

▶ Show code

stat
<dbl>
130.3519

1 row

▶ Show code

p_value
<dbl>
0.008

1 row

Explanation & conclusion: We first compute the observed mean revision. Then we pretend the true mean revision is 0 and, using the bootstrap, create 2,000 resampled datasets and record the mean for each. get_p_value() compares our observed mean to this simulated distribution and returns a two-sided p-value.

In your write-up, interpret the number in boot_p_mean$p_value: If it's small (e.g., < 0.05), you can say there is bootstrap-based evidence that the average CES revision is not zero. If it's large, you can say the bootstrap procedure suggests that the mean revision could plausibly be zero given the noise in the data.

D. Bootstrap test 2 — Is the median revision different from zero?

Here we do a bootstrap test for the median, which does not have a simple closed-form test like the t-test. This is similar in spirit to a Wilcoxon-type test.

▶ Show code

stat
<dbl>
482

1 row

▶ Show code

p_value
<dbl>
0

1 row

Explanation & conclusion: - We compute the observed median revision. - Under the null hypothesis that the true median is 0, we bootstrap to get 2,000 possible median values. - The resulting p-value tells us whether our observed median is unusually far from zero compared with what we'd see by chance. In your narrative, compare this p-value to the one from the mean test; you can comment on whether revisions tend to be slightly positive or negative on average, and whether this is robust when you focus on the median (which is less sensitive to outliers).

E. Bootstrap test 3 — Is the probability of a negative revision equal to 50%?

This is a bootstrap analogue of a binomial proportion test.

▶ Show code

stat
<dbl>
0.2495512

1 row

▶ Show code

p_value
<dbl>
1

1 row

Explanation & conclusion: - We convert revisions into a TRUE/FALSE variable: TRUE when the revision is negative. - We ask whether the probability of a negative revision is exactly 50%. - Under that null, we bootstrap the proportion 2,000 times and compare our observed proportion to this distribution. In your write-up, explain whether the p-value suggests that negative revisions are more or less common than positive ones, and how strong that evidence is.

F. Schematic "how-to" visualization (conceptual flowchart)

You can add a very simple "flowchart" using plain markdown (no code), which satisfies the requirement that it is not just a ggplot of a sampling distribution.

For example:

How a bootstrap test works (in words)

1. Start with your sample (all CES revisions).
2. Define the null hypothesis (e.g., "true mean revision = 0").
3. Resample with replacement from the data to create a new "bootstrap" dataset.
4. Recalculate the statistic (mean, median, or proportion).
5. Repeat steps 3–4 thousands of times to build a distribution of the statistic under the null.
6. Locate the observed statistic on this bootstrap distribution. The p-value is the fraction of bootstrap statistics that are at 7. least as extreme as what we observed.
7. Use this p-value to decide whether the data provide strong evidence against the null.